TRAINING AND CHECKPOINT SELECTION

The reward and checkpoints are selected with development routes before the final evaluations.

1. Staged reward search
Screen 7 handover costs on 6 development routes. Keep candidates passing the A3-style throughput and mobility gate.

2. Train three policies
Fix c_H = 0.875; use seeds 17, 27, and 37. Train on the same 300-journey plan.

3. Select checkpoints
Check saved checkpoints on the same six routes. Apply the full throughput and mobility gate.

4. Freeze and evaluate
Freeze one checkpoint per seed. Test greedily on reserved and Pokhara cases; no learning.

Inside each training run

Observe
Cell features, graph, legal-action mask

Choose action
Online Q network; ε-greedy choice from legal cells

Run simulator
Record reward, duration, next state, and action mask

Replay and update
Double-DQN update from a batch of stored transitions

repeat until the 300-journey plan is complete

Training settings

ε: 1 for 2,000 steps, then linear to 0.05 over 100,000 | Replay capacity: 20,000 | Batch: 32 | Adam: 0.001 | Huber loss | Gradient clip: 10
Update every 4 environment steps after warm-up; copy target-network weights every 1,000 optimizer updates.

Final evaluation: 639 reserved route cases and 540 Pokhara transfer groups. Checkpoints are not chosen from these outcomes.